

modify policies, allowing tourists to use satellite phones in the Himalayas and other regions of India, including parts of Arunachal Pradesh, where conventional network connectivity is not good. This will provide a further boost for satellite communications in the country.

On the device front too, satellite phones are now much better than they were a few years ago. Modern instruments can withstand extreme temperatures, harsh downpours and even terrible falls. Their communication capability is also much higher.

The complaints people usually have against satellite phones are that they are a bit chunky and also do not offer many value-added services and apps. However, satellite service provider Globalstar has recently offered an interesting service called Sat-Fi, which creates a satellite hotspot for any Wi-Fi enabled device. This can essentially turn any smartphone, tablet or computer into a satellite phone! So, you can enjoy all the conveniences of a smartphone at normal times and resort to satellite-based voice and data connectivity when cellular service is not available in the vicinity.

The coldest corners too

Just as the denseness of the Amazons deters service providers, so does the emptiness of the Arctic, making explorers—and defence organisations—to seek alternate means of communications. As one of the effects of global warming, a lot of sea lanes are getting opened up in the summer, which offers not only more opportunity to explore but security risks as well. In such a situation, better communication is highly essential not just to keep in touch and coordinate with others but also to use/develop maps and such applications. This

Google's Balloons

Reading about communications in godforsaken lands and unexplored frontiers might make some wonder how much better it would be if some of those efforts were pumped into providing connectivity to the remote villages of India, where sometimes people have to climb a tree or walk for kilometres to get a signal on their mobile phone. Well, that is the trouble Google's Project Loon aims to shoo away. Balloons fitted with communication infrastructure, floating in the stratosphere at around 18.29km (60,000 feet) above the ground, are being used by a team at Google to provide high-speed Internet to schools and others in remote regions of the world. In order to connect a location to Project Loon, Google installs another small balloon with a special antenna on the building, which enables them to connect to the high-flying balloon network, providing high-speed Internet access. The project has been successfully tested in remote parts of New Zealand, California and Brazil. Why not try and bring them to your region too? Read more about Google's Project Loon, the Balloon Powered Internet for Everyone, at <http://www.google.co.in/loon/>.



The Loon infrastructure is launched 18.29km (60,000 feet) above the ground because the stratosphere, which has no clouds to block the sun, is the perfect place for solar-powered equipment (Courtesy: Google)

has made the US Navy and the Coast Guard to explore means to improve the communications architecture there.

Like in the mountains and other unreachable areas, satellite communication is the best option here too—and has been used since many years. However, the legacy satellite infrastructure, the UHF Follow-On, was found to be unreliable in latitudes higher than 65 degrees north. As these satellites are in geosynchronous orbits above the Equator, their beams cover most of the Earth, but fade in the polar regions.

In the near future, the Navy is betting on its mobile user objective system (MUOS) and the commercial satellite system Iridium, which is set to launch its next-generation fleet of spacecraft in 2015. Two of the MUOS satellites are already in orbit, and two will be up soon. These MUOS satellites will have more beams, more power and a waveform based on 3G-wideband-spread spectrum. Therefore these will be capable of bending around the curvature

of the Earth. The reach of the current satellites has been tested well above 80 degrees now. They provide 24-hour coverage of the entire Northwest Passage. Although the communication used is narrow-band, they were able to send decent large data files of around 289MB. Once the remaining satellites are also ready, there will be a seamless handover from one spacecraft to another as one satellite rises in orbit and the other falls.

When their new satellite system is up, Iridium is expected to be of great help, supplementing the coverage of the MUOS satellites and extending connectivity to explorers and civilians too. Iridium's 66 satellites will operate in low-Earth orbit, ensuring that every point on the planet is covered by at least one satellite. That means you and I will be crossed by their beams too, wherever we are.

Well, magical beams, invisible signals, bright balloons, unheard acoustics, all these wonderful communication technologies ensure we are always within reach of help. Should we be in danger, we can be in touch with anybody from anywhere and access any information whenever and wherever we want. The world is in our hands—literally—whether we are deep within the bellies of the Earth or high up in space. What was once the dream of kindergarten kids later became the realm of scientists and the play field of industrialists, only to become reality today! ●

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Satellites to the rescue

Satellite communication can come to the rescue not just in harsh terrains but also in disaster-struck regions. Recently, Mitsubishi Electric Corporation developed the world's first helicopter satellite communication system (HSA40) for use by Japan's Fire and Disaster Management Agency (FDMA). The system has been deployed to gather information from disaster-struck areas, so that relief measures can be rapidly deployed. Earlier, the FDMA used a helicopter fitted with a Heli-TV system, which transmitted TV signals from a helicopter to a relay station on the ground. This system, however, was prone to transmission disruptions by mountains and buildings and was unusable in the absence of a relay station on the ground. HSA40, on the other hand, uses satellite communication. It transmits video and voice data from the helicopter to a satellite, enabling real-time information such as aerial video to be transmitted reliably to base stations anywhere in Japan. This system does not require any relay stations for transmission and also enables bi-directional communication. It is very lightweight, ideal for being airborne.